SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 20%

Duration : 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Reverse Engineering

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Reverse Engineering

1. System Decomposition & Analysis

Analyze an existing AI-based system and **reverse engineer its architecture**, identifying key components, data flow, and underlying algorithms.

2. Architecture Reconstruction

Given a black-box software system, **reconstruct its system architecture** and explain the interaction between modules using appropriate diagrams.

3. Algorithm Identification

Reverse engineer a machine learning application to **identify the algorithms, data processing steps, and decision-making logic** used in the system.

4. Data Flow & Process Mapping

Examine an existing distributed AI application and **map its data flow, processing pipeline, and communication between components**.

5. Improvement & Redesign

Perform reverse engineering on an intelligent system and **propose enhancements or redesign strategies** to improve scalability, efficiency, and performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation